OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

STATION: Salem Generating Station

SYSTEM: Main Turbine

TASK: Perform Main Turbine Valve Testing

TASK NUMBER: N0450130201

JPM NUMBER: 14-01 NRC Sim d

ALTERNATE PATH: [X]

K/A NUMBER: 045 2.1.23

IMPORTANCE FACTOR: 4.3 (RO) 4.4 (SRO)

APPLICABILITY:
EO [ ] RO [X] STA [ ] SRO [X]

EVALUATION SETTING/METHOD: Simulator / Perform

REFERENCES: S2.OP-PT.TRB-0003, Rev. 19 (checked 9-23-15)

TOOLS AND EQUIPMENT: None

VALIDATED JPM COMPLETION TIME: 15 minutes

TIME PERIOD IDENTIFIED FOR TIME CRITICAL STEPS: N/A

Developed By: **G Gauding** **Instructor** **Date: 8-20-15**

Validated By: **S Bickhart / J Militti** **SME or Instructor** **Date: 9-23-15**

Approved By: R Williams **Training Department** **Date:** 10-7-15

Approved By: M06 **Operations Department** **Date:** 10915

ACTUAL JPM COMPLETION TIME:

ACTUAL TIME CRITICAL COMPLETION TIME:

PERFORMED BY:

GRADE: [ ] **SAT** [ ] **UNSAT**

REASON, IF UNSATISFACTORY:

EVALUATOR'S SIGNATURE: **DATE:**

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: ______________________

DATE: ______________________

SYSTEM: Main Turbine

TASK: Perform Main Turbine Valve Testing

TASK #: N0450130201

Simulator Setup: **IC-244 22RS5 fails shut after 22RS5 is shut during test. MALF: VL0662 Reheat Intercept R2E fails to position (0-100%) Severity 0, tied to ET-1: tmvrs06l(2)**

INITIAL CONDITIONS:

Unit 2 is operating at 89% power, MOL. Power is reduced to perform Main Turbine Valve Testing. Rod control is in manual per CRS direction to prevent rod movement during testing, with Control Bank D at 187 steps. The ESO has been notified of valve test.

INITIATING CUE:

Perform Main Turbine Valve Testing IAW S2.OP-PT.TRB-0003, Main Turbine Valve Stroke Testing, Section 5.3 Main Turbine Valve Stroke Testing-Turbine Operating OR Main Steam Isolation Valves Closed, starting at Step 5.3.6 to perform Reheat and Intercept valve testing. Section 5.2, Test Preparation has been completed. Personnel are at the Main Turbine to perform any required local actions.

Successful Completion Criteria
1. All critical steps completed. 2. All sequential steps completed in order. 3. All time-critical steps completed within allotted time. 4. JPM completed within validated time. Completion time may exceed the validated time if satisfactory progress is being made

Task Standard for Successful Completion:
1. Perform stroke time testing of 21RS5 and 21RS6 SAT. **2. Initiate a power reduction at 10% per hour within 5 minutes of the 22RS5 being shut during stroke time testing.**

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: ______________________

DATE: ______________________

SYSTEM: Main Turbine

TASK: Perform Main Turbine Valve Testing

*	STEP No.	STEP * Denotes critical task	STANDARD	EVAL S/U	COMMENTS
			Note to Evaluators: Valve positions are required to be checked LOCALLY to satisfy test requirements. IF **only** EHC console indication is used, then JPM should be considered a failure, as it would result in exceeding periodicity FSAR requirements for turbine valve testing (10.2.2.6, page 10.2-7)		
		Provide marked up copy of S2.OP-PT.TRB-0003, Main Turbine Valve Stroke Testing			
	5.3.1	IF the Main Turbine is shutdown, THEN:	Determines Main Turbine is operating.		
	5.3.2	IF performing 21MS28/21MS29 testing, THEN:	Determines MS28 and MS29 testing will be performed after reheat valve testing is performed.		
	5.3.6	IF testing 21RS5 - EAST REHEAT STOP AND 21RS6 - EAST INTERCEPT, THEN:			
*	5.3.6.A	At the TURBINE E-H CONTROL & STATUS monitor, EAST LP VALVE TESTS	Selects EAST LP VALVE TESTS screen, and determines the TEST PERMISSIVE, NO		

*	STEP No.	STEP * Denotes critical task	STANDARD	EVAL S/U	COMMENTS
		screen, **ENSURE** the TEST PERMISSIVE, NO OTHER TESTS IN PROGRESS is green.	OTHER TESTS IN PROGRESS is green.		
	5.3.6.B	**DIRECT** an Operator to locally monitor 21RS5 AND 21RS6 for full stroke.	Directs field operator monitor 21RS5 AND 21RS6 for full stroke. **Cue: I am standing by to monitor 21RS5 and 21RS6 for full stroke.**		
	5.3.6.C	**RECORD** 21RS5 AND 21RS6 positions on Attachment 2, Section 3.0, by initialing TEST POSITION 1.	Records **<u>local</u>** 21RS5 and 21RS6 positions on Attachment 2, Section 3.0, by initialing TEST POSITION 1. (open) **Cue when asked: 21RS5 and 21RS6 are fully open.**		
		<u>NOTE</u> The MicroNet System will perform the following functions during testing: Issue an alarm for any valve out-of-position during testing Energize 21RS5 and 21RS6 solenoids to CLOSE both valves Hold each valve in the closed position for 10 seconds after both valves are detected in the CLOSED position De-energize 21RS5 and 21RS6 solenoids after 10 seconds elapses Alarm and initiate ABORT TESTING after 1			

*	STEP No.	STEP * Denotes critical task	STANDARD	EVAL S/U	COMMENTS
		minute of TEST IN PROGRESS			
*	5.3.6.D	At the TURBINE E-H CONTROL & STATUS monitor, 1. **SELECT** START TEST for 21RS5 AND 21RS6. 2. **ENSURE** NORMAL OPERATION changes to TEST IN PROGRESS.	At the TURBINE E-H CONTROL & STATUS monitor, selects START TEST for 21RS5 AND 21RS6 and determines NORMAL OPERATION changes to TEST IN PROGRESS.		
	5.3.6.E	When 21RS5 AND 21RS6 OPEN/MOVING indications are cleared, THEN **RECORD** 21RS5 AND 21RS6 positions on Attachment 2, Section 3.0, by initialing TEST POSITION 2.	When 21RS5 AND 21RS6 OPEN/MOVING indications are cleared, records 21RS5 AND 21RS6 local positions on Attachment 2, Section 3.0, by initialing TEST POSITION 2. (closed) **Cue when asked: 21RS5 and 21RS6 went closed.**		
		<u>CAUTION</u> **Turbine-Generator load must be reduced to less than 80% at 10%/hr when a Reheat Stop Valve or Intercept Valve fails to reopen within 5 minutes while above 80% load.**			
	5.3.6.F	When TEST IN PROGRESS changes to NORMAL OPERATION, THEN **RECORD** 21RS5 AND 21RS6 positions on Attachment 2, Section 3.0, by initialing TEST POSITION 3.	When test in progress changes to normal operation, records 21RS5 and 21RS6 local positions on Attachment 2, Section 3.0 by initialing TEST POSITION 3. (open) **Cue when asked: 21RS5 and 21RS6 are fully**		

*	STEP No.	STEP * Denotes critical task	STANDARD	EVAL S/U	COMMENTS
			open.		
	5.3.6.G	IF EAST LP VALVE TESTS is completed, THEN **SELECT** CLOSE WINDOW at the TURBINE E-H CONTROL & STATUS monitor.	Determines testing is not complete.		
	5.3.7	IF testing 22RS5 - EAST REHEAT STOP AND 22RS6 - EAST INTERCEPT, THEN:			
	5.3.7.A	At the TURBINE E-H CONTROL & STATUS monitor, EAST LP VALVE TESTS screen, **ENSURE** the TEST PERMISSIVE, NO OTHER TESTS IN PROGRESS is green.	Observes already on EAST LP VALVE TESTS screen, and determines the TEST PERMISSIVE, NO OTHER TESTS IN PROGRESS is green.		
	5.3.7.B	**DIRECT** an Operator to locally monitor 22RS5 AND 22RS6 for full stroke.	Directs field operator monitor 22RS5 AND 22RS6 for full stroke. **Cue: I am standing by to monitor 22RS5 and 22RS6 for full stroke.**		
	5.3.7.C	**RECORD** 22RS5 AND 22RS6 positions on Attachment 2, Section 3.0, by initialing TEST POSITION 1.	Records **local** 22RS5 and 22RS6 positions on Attachmen1, Section 3.0, by initialing TEST POSITION 1. (open) **Cue when asked: 22RS5 and 22RS6 are fully open.**		
		NOTE The MicroNet System will perform the			

*	STEP No.	STEP * Denotes critical task	STANDARD	EVAL S/U	COMMENTS
		following functions during testing: Issue an alarm for any valve out-of-position during testing Energize 22RS5 and 22RS6 solenoids to CLOSE both valves Hold each valve in the closed position for 10 seconds after both valves are detected in the CLOSED position De-energize 22RS5 and 22RS6 solenoids after 10 seconds elapses Alarm and initiate ABORT TESTING after 1 minute of TEST IN PROGRESS			
*	5.3.7.D	At the TURBINE E-H CONTROL & STATUS monitor, 1. **SELECT** START TEST for 22RS5 AND 22RS6. 2. **ENSURE** NORMAL OPERATION changes to TEST IN PROGRESS.	At the TURBINE E-H CONTROL & STATUS monitor, selects START TEST for 22RS5 AND 22RS6 and determines NORMAL OPERATION changes to TEST IN PROGRESS. **EVALUATOR MARK TIME 22RS5 indicates shut:** ________:________:________		
	5.3.7.E	When 22RS5 AND 22RS6 OPEN/MOVING indications are cleared, THEN **RECORD** 22RS5 AND 22RS6	When 22RS5 AND 22RS6 OPEN/MOVING indications are cleared, records 22RS5 AND 22RS6 local positions on Attachment 2,		

*	STEP No.	STEP * Denotes critical task	STANDARD	EVAL S/U	COMMENTS
		positions on Attachment 2, Section 3.0, by initialing TEST POSITION 2.	Section 3.0, by initialing TEST POSITION 2. (closed)		
			Note: After 10 seconds of being closed, 22RS6 will re-open, and 22RS5 remains shut.		
		<u>CAUTION</u> **Turbine-Generator load must be reduced to less than 80% at 10%/hr when a Reheat Stop Valve or Intercept Valve fails to reopen within 5 minutes while above 80% load.**			
			Determines 22RS5 has not opened, and contacts local operator to verify position. **Cue: 22RS6 is open and 22RS5 is shut.**		
	5.2.3	IF any valve does not respond as expected or fails to reopen during this test, THEN **PERFORM** the following troubleshooting before continuing on with the test: A. **CHECK** fuse LED for any lit LEDs (failed fuse) AND **REPLACE** fuse, if needed. B. **CYCLE** the associated EHC valve to attempt to reopen the valve. C. **CHECK** associated Limit Switches (HMI/Locally). D. Lightly **AGITATE** the associated solenoid valve with a small hammer. E. **OPEN** associated fuse holder, reinstall	**Note: This step is contained in Section 5.2, TEST PREPARATION.** Contacts field operator to investigate 22RS5 malfunction IAW Step 5.2.3. **Note: Other than reporting 22RS5 position is closed, do not provide any other report or respond to plant page after giving report of valve position requested at this step.**		

*	STEP No.	STEP * Denotes critical task	STANDARD	EVAL S/U	COMMENTS
		fuse, and **CLOSE** fuse holder. F. IF the valve functions following any of the above steps, THEN **INITIATE** a Notification to document any alarms on the HMI screen and what caused the valve to respond. G. IF none of the above step resolve the problem, THEN **CONTACT** I&C to perform additional troubleshooting.			
*			Within 5 minutes of time marked above at step 5.3.7.D, initiates a turbine load reduction at 10% per hour. **Cue as required that RO will initiate boration and maintain rod control.** **If asked if the CRS wants a load reduction performed, respond: "The CRS states that if a load reduction is required by procedure to initiate the load reduction as per procedural direction."**		
			Terminate the JPM once a load reduction at 10% per hour is commenced OR 5 minutes has elapsed from the time 22RS5 was shut.		

TERMINATING CUE: None

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

JOB PERFORMANCE MEASURE VALIDATION CHECKLIST

NOTE: All steps of this checklist should be performed upon initial validation. Prior to JPM usage, revalidate JPM using steps 8 and 11 below.

QS 1. Task description and number, JPM description and number are identified.

QS 2. Knowledge and Abilities (K/A) references are included.

QS 3. Performance location specified. (in-plant, control room, or simulator)

QS 4. Initial setup conditions are identified.

QS 5. Initiating and terminating Cues are properly identified.

QS 6. Task standards identified and verified by SME review.

QS 7. Critical steps meet the criteria for critical steps and are identified with an asterisk (*).

QS 8. Verify the procedure referenced by this JPM matches the most current revision of that procedure: Procedure Rev. 19 Date 5/27/2015

QS 9. Pilot test the JPM:
   a. verify Cues both verbal and visual are free of conflict, and
   b. ensure performance time is accurate.

________ 10. If the JPM cannot be performed as written with proper responses, then revise the JPM.

________ 11. When JPM is revalidated, SME or Instructor sign and date JPM cover page.

SME/Instructor: [signature] Rickhart Date: 9/23/15

SME/Instructor: [signature] Milith Date: 9-23-15

SME/Instructor: ____________________ Date: __________

INITIAL CONDITIONS:

Unit 2 is operating at 89% power, MOL. Power is reduced to perform Main Turbine Valve Testing. Rod control is in manual per CRS direction to prevent rod movement during testing, with Control Bank D at 187 steps. The ESO has been notified of valve test.

INITIATING CUE:

Perform Main Turbine Valve Testing IAW S2.OP-PT.TRB-0003, Main Turbine Valve Stroke Testing, Section 5.3 Main Turbine Valve Stroke Testing-Turbine Operating OR Main Steam Isolation Valves Closed, starting at Step 5.3.6 to perform Reheat and Intercept valve testing. Section 5.2, Test Preparation has been completed. Personnel are at the Main Turbine to perform any required local actions.